PHYSICAL CHEMISTRY CUMULATIVE EXAM
October 22, 1999

SOLUBILITY PRODUCTS & ION ACTIVITIES

What is the concentration of Ca^{2+} ions in an aqueous solution that is saturated with calcite at 25°C? The change in Gibbs function for the dissolution is +47.40 kJ when the reaction is written:

$$CaCO_3(s,calcite) \text{ <-----> } Ca^{2+}(aq) + CO_3^{2-}(aq)$$

(For the following, you may disregard the autoprotolysis of water)

First, find the calcium ion concentration in the solution, assuming ideal behavior for all the ions present:

Now consider that the ions themselves alter the ionic strength, and hence the activities of the product ions. Use the Debye-Huckel limiting law to find the new concentration of calcium ions:

Next, consider the equilibrium to take place in a solution that is initially 0.010 mol kg^{-1} in Na_2CO_3 (i.e., the "common-ion effect"). Predict the Ca^{2+} concentration in this case:

Finally, consider the equilibrium in the presence of 0.010 mol kg^{-1} of the inert salt $NaNO_3$ (i.e., the "diverse-ion effect"). Predict the Ca^{2+} concentration:

Debye-Hückel Limiting Law: $\lg \gamma_{\pm} = -|z_+z_-| A I^{1/2}$

where 'lg' is the base-10 logarithm, $\gamma_{\pm}$ the mean activity coefficient, $z_{+(-)}$ the ion charge numbers, I the ionic strength, and A = 0.509 for aqueous solutions at 25°C.